Unit 3 - Normalization (CS301 Database Management Systems)

Normalization organizes a relational schema to reduce redundancy and avoid update, insert and delete anomalies. We decompose a relation into smaller relations using functional dependencies (FDs).

A functional dependency X -> Y means that the value of X determines Y.
Example: in Student(RollNo, Name, Dept, HOD), RollNo -> Name, Dept and Dept -> HOD.

First Normal Form (1NF): every attribute holds a single atomic value.
A column storing a list of phone numbers violates 1NF; move phones to their own table StudentPhone(RollNo, Phone).

Second Normal Form (2NF)

A relation is in 2NF if it is in 1NF and no non-key attribute depends on only part of a composite key (no partial dependency).
Example: Enrollment(RollNo, CourseID, StudentName, Grade). StudentName depends only on RollNo, so split into Student(RollNo, StudentName) and Enrollment(RollNo, CourseID, Grade).

Third Normal Form (3NF)

A relation is in 3NF if it is in 2NF and no non-key attribute depends on another non-key attribute (no transitive dependency).

Worked example: Student(RollNo, Name, Dept, HOD) with RollNo -> Dept and Dept -> HOD. HOD depends on RollNo only through Dept, a transitive dependency. If a department changes its HOD we must update many rows (update anomaly).
Decompose into Student(RollNo, Name, Dept) and Department(Dept, HOD).
Both relations are now in 3NF and the join on Dept is lossless.

BCNF is a stricter form: for every FD X -> Y, X must be a superkey.